

Gaze-Glide: Enhanced Eye-Gaze Controlled Wheelchair with GPS Integration for Improved Mobility.

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Abstract : Gaze Glide is an innovative eye-tracking control system designed to enhance mobility and independence for individuals with ALS and other motor impairments. This technology enables hands-free navigation of powered wheelchairs through precise eye gaze input, offering an intuitive and seamless user experience. By leveraging advanced eye-tracking algorithms and integrating them with a Raspberry Pi-based control system, Gaze Glide ensures accurate directional control and real-time responsiveness..

Keywords: Eye-gaze tracking, Raspberry pi, Open CV, GPS integration, Intelligent Wheelchair, ALS, Beep Learning, Mobility Assistance, IR Camera.

I. INTRODUCTION

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that significantly limits motor functions while often leaving cognitive abilities intact. This condition creates an urgent need for assistive technologies that enable mobility and communication for individuals with ALS. Traditional mobility aids, such as manual or powered wheelchairs, often fall short in addressing the needs of ALS patients, who may lack control over their limbs or voice. This project builds upon prior research on eye-gaze-controlled wheelchairs by incorporating a GPS module to enhance navigation capabilities. The combination of eye-gaze tracking powered by deep learning and GPS technology enables a more comprehensive solution for both indoor and outdoor mobility. This system offers not only increased independence for ALS patients but also potential applications for other users with mobility impairments. Individuals with ALS, spinal cord injuries, and other neuromuscular disorders often face significant mobility challenges, limiting their independence and quality of life. Traditional powered wheelchairs rely on joystick or switch-based controls, which may become difficult or impossible to operate as motor function deteriorates.

To address this issue, Gaze Glide introduces an advanced eye-tracking control system that enables hands-free wheelchair navigation through intuitive gaze input.

Gaze Glide leverages cutting-edge eye-tracking technology integrated with a Raspberry Pi-based control unit to interpret users' eye movements and translate them into precise directional commands. This system offers an accessible and efficient alternative to conventional control methods, empowering individuals with severe mobility impairments to navigate their environment with greater ease. Additionally, it incorporates intelligent turning mechanisms and obstacle detection to enhance safety and maneuverability, ensuring smooth and responsive navigation in dynamic settings.

II. LITERATURE REVIEW

Assistive technology has played a crucial role in enhancing mobility for individuals with neuromuscular disorders such as ALS, cerebral palsy, and spinal cord injuries. Traditional powered wheelchairs commonly rely on joystick, head switches, or sip-and-puff systems to enable movement (Kim et al., 2020). However, these methods become challenging or unusable as motor impairments progress, necessitating alternative control solutions. Recent advancements in human-computer interaction have led to the development of eye-tracking technology, providing a hands-free approach to device control (Majaranta & R  ih  , 2018). Eye-tracking has been widely studied for communication purposes, particularly in augmentative and alternative communication (AAC) devices, and has shown potential for integration with mobility systems (Haji et al., 2021).

Several studies have explored the feasibility of gaze-based wheelchair control. Arai et al. (2019) successfully demonstrated an eye-controlled wheelchair that responded to gaze direction, improving user autonomy. However, challenges such as calibration errors, slow response times, and unintended movements due to involuntary eye motions remain significant barriers (Gonzalez et al., 2022). To address these limitations, researchers have proposed intelligent filtering algorithms, improved gaze-detection accuracy, and multimodal control approaches that combine eye tracking with voice commands or facial gestures (Sharma et al., 2021).

Obstacle detection and path planning are also critical for safe navigation. Studies suggest that integrating real-time obstacle avoidance systems with gaze control can improve maneuverability in dynamic environments (Li et al., 2020). Machine learning techniques have been explored to enhance predictive movement and adaptability, further refining the accuracy of gaze-based control systems (Chen et al., 2021).

Gaze Glide builds upon this existing research by integrating real-time gaze tracking with a Raspberry Pi-based control system, incorporating intelligent turning controls and obstacle avoidance. By addressing key challenges in accuracy and responsiveness, Gaze Glide aims to offer a reliable, efficient, and accessible solution for individuals with severe mobility impairments

III. METHODOLOGY

System Architecture :

The development of Gaze Glide involves the integration of eye-tracking technology with a Raspberry Pi-based control system to enable hands-free wheelchair navigation. The methodology follows a structured approach, including system design, hardware integration, software development, and performance evaluation.

System Design and Hardware Integration :

The system consists of an eye-tracking module, a Raspberry Pi microcontroller, motor controllers, and ultrasonic sensors for obstacle detection. The eye-tracking device captures gaze input and translates it into directional commands. The Raspberry Pi processes these commands and communicates with the wheelchair's motor controller to execute movement. Ultrasonic sensors provide real-time obstacle detection, ensuring safe navigation.

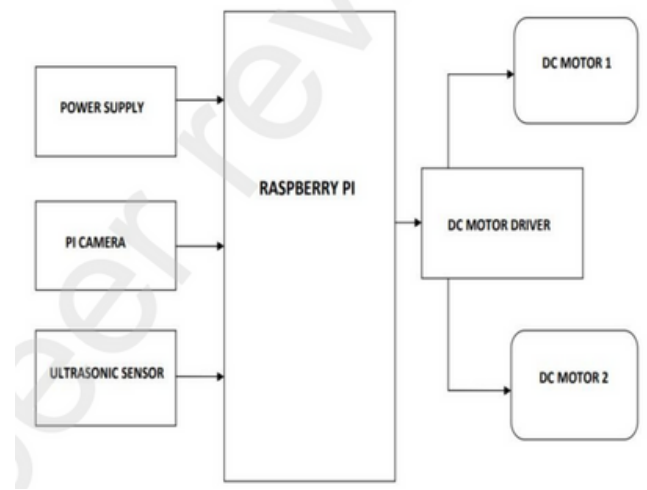


fig 1: System Architecture

Software Development :

A Python-based algorithm interprets gaze data and converts it into movement commands. A filtering mechanism is implemented to reduce errors caused by involuntary eye movements. The system also includes a calibration module to personalize settings for each user.

Performance Evaluation :

Testing is conducted in a controlled environment to assess response time, accuracy, and user comfort. Metrics such as gaze detection reliability, movement precision, and obstacle avoidance efficiency are analyzed. User feedback is collected to refine the system's usability.

This structured methodology ensures that Gaze Glide is both functional and user-friendly, improving accessibility for individuals with severe mobility impairments.

GPS Integration :

The GPS module uses a high-precision receiver to track the wheelchair's location. A navigation algorithm calculates optimal routes based on user-defined destinations, considering accessibility features like ramps and wide pathways. The integration of GPS and eye-gaze control allows users to select and approve routes interactively.

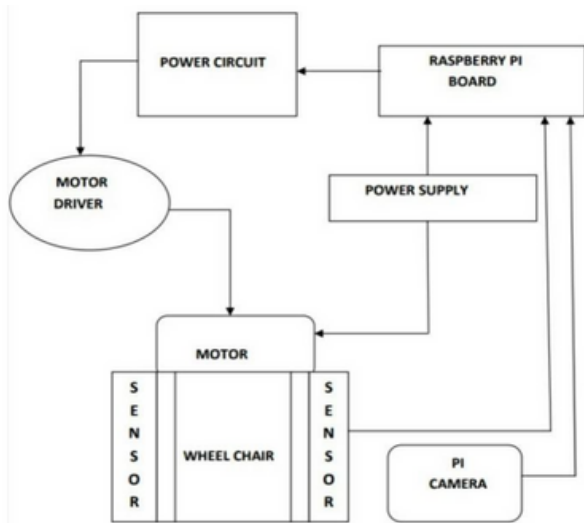


fig 2: Design Model

IV. IMPLEMENTATION

The implementation of Gaze Glide involves the seamless integration of eye-tracking technology with a Raspberry Pi-based control system, enabling hands-free wheelchair navigation. The process follows a structured approach that includes hardware integration, software development, system calibration, obstacle detection, and user testing.

Hardware integration :

The system comprises an eye-tracking device, a Raspberry Pi microcontroller, motor drivers, and an infrared (IR) camera for night vision and obstacle detection. The eye-tracking device captures gaze direction and transmits data to the Raspberry Pi, which processes the input and generates movement commands for the wheelchair. The IR camera ensures real-time object detection and environmental awareness, providing enhanced safety in both daylight and low-light conditions. Unlike ultrasonic sensors, which rely on sound waves, the IR camera offers improved detection accuracy in varied lighting environments.

Software Development :

The software, developed primarily in Python, processes gaze-based inputs and converts them into directional movement commands such as forward, left, right, and stop. A gaze stabilization algorithm minimizes errors caused by involuntary eye movements, ensuring smooth navigation. The IR camera feed is processed using computer vision techniques, such as edge detection and depth mapping, to identify obstacles. Machine learning models can be integrated in future iterations to improve obstacle recognition accuracy. A graphical user interface (GUI) allows users to customize sensitivity settings and recalibrate the system as needed.

V. APPLICATION AND FUTURE WORK

Gaze Glide has significant applications in assistive mobility, particularly for individuals with ALS, spinal cord injuries, and other neuromuscular disorders that limit motor function. By providing hands-free wheelchair control through eye-tracking technology, the system enhances independence and accessibility for users who cannot operate traditional input devices like joysticks or switches. Additionally, its integration with an infrared (IR) camera for night vision and obstacle detection ensures safe navigation in various lighting conditions, making it suitable for both indoor and outdoor use. Beyond wheelchair control, the technology could be adapted for other assistive applications, such as smart home control, robotic navigation, and rehabilitation systems.

Future work will focus on improving system accuracy and adaptability. Enhancing the gaze-tracking algorithm to better filter involuntary eye movements will reduce unintentional commands and improve user experience. Further research into machine learning-based predictive movement models could allow the system to anticipate user intentions, making navigation smoother and more intuitive. Expanding the system's compatibility with different eye-tracking hardware and incorporating voice or gesture-based hybrid controls could enhance accessibility for a wider range of users. Extensive user trials and real-world testing will be conducted to refine usability and reliability, ensuring that Gaze Glide continues to evolve as an innovative assistive mobility solution.

VI. CONCLUSION

Gaze Glide represents a significant advancement in assistive mobility technology, providing a hands-free, intuitive solution for individuals with severe physical disabilities. By integrating eye-tracking technology with a Raspberry Pi-based control system, Gaze Glide enables users to navigate a powered wheelchair using only their gaze, eliminating the need for traditional joystick or switch-based controls. The inclusion of an infrared (IR) camera for obstacle detection and night vision enhances safety, allowing for reliable mobility in various lighting conditions.

The development and implementation of Gaze Glide highlight the potential of eye-tracking as a viable alternative for mobility control. The system's gaze stabilization algorithm minimizes errors caused by involuntary eye movements, while the real-time processing of gaze input ensures smooth and responsive navigation.

VII. REFERENCES

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